

AZUBUIKE PRECIOUS

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EXECUTIVE SUMMARY

Results-driven Artificial Intelligence and Computer Science graduate with hands-on experience designing and implementing AI-driven solutions, including conversational AI systems, automation workflows, and computer vision applications. Passionate about exploring emerging AI technologies and rapidly translating business problems into practical Proof of Concepts (POCs) and intelligent solutions. Experienced in building AI-powered applications using Python, APIs, machine learning, and NLP techniques, with growing exposure to Generative AI, LLM integrations, and agentic AI concepts. Skilled in evaluating user requirements, designing scalable workflows, integrating AI services, and delivering functional prototypes that improve operational efficiency and user experience. Strong ability to bridge the gap between business stakeholders and technical implementation by translating ideas into practical AI-driven solutions. Comfortable working in fast-paced, innovation-focused environments, collaborating across teams, and quickly adapting to unfamiliar technologies and frameworks. Currently expanding expertise in Generative AI ecosystems, including LLM orchestration, AI automation frameworks, prompt-driven workflows, and production-ready AI solution design.

SKILLS AND CORE COMPETENCIES

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| ▪ Generative AI and LLM Applications | ▪ Data Analysis and Data-Driven Decision-Making |
| ▪ AI Solutions Prototyping and POC Development | ▪ Cloud Computing Fundamentals (AWS / Azure) |
| ▪ Python Programming Automation | ▪ Solution Architecture and Cloud Architecture |
| ▪ API Integration and AI Service Integration | ▪ Model deployment and optimization |
| ▪ Conversation AI and Chatbot Development | ▪ Problem Solving & Analytical Thinking |
| ▪ Prompt Engineering & Workflow Design | ▪ Testing, Debugging & Quality Assurance |
| ▪ Agentic AI Concepts & AI Orchestration | ▪ REST API and Back end integration |
| ▪ NLP Concepts & Text Processing | ▪ Agile & Collaborative Development (Scrum & Kanban) |
| ▪ LangChain & LLM Workflow Fundamentals | ▪ Collaboration Across Cross-Functional Teams |
| ▪ Machine Learning & computer vision | |

WORK EXPERIENCE

AI Project Developer | June 2024 – Date

Hertfordshire University

Helmet Detection System (CNN/SVM)

Developed a computer vision model for detecting helmets on roads and construction sites.

Key Responsibilities

- Designed and developed an AI-based computer vision solution to detect helmet usage across road and construction site environments
- Built and trained machine learning models using Convolutional Neural Networks (CNN) and Support Vector Machines (SVM)

- Collected, cleaned, and labelled image datasets to support supervised learning and improve model performance
- Evaluated and optimised model accuracy using performance metrics, tuning parameters to enhance detection reliability
- Collaborated with academic staff and stakeholders to align the solution with real-world safety monitoring requirements
- Documented system design, methodology, and outcomes to support knowledge sharing and future development

Key Achievements:

- Delivered a fully functional helmet detection system capable of identifying safety compliance in visual data
- Improved model performance through effective data pre-processing and iterative model tuning
- Demonstrated practical application of AI in enhancing health and safety monitoring
- Successfully delivered the solution within agreed timelines, meeting project and stakeholder expectations

Machine Learning Engineer | January 2023 – May 2024

DRE Automation Solution

Birmingham

AI Chatbot with Email Automation

Developed an intelligent AI-powered chatbot system designed to automate user communication workflows and improve operational efficiency through automated responses and API driven integrations.

Key Responsibilities

- Designed and developed a conversational AI solution capable of processing user requests and triggering automated communication workflows
- Implemented backend logic and integrated APIs to support automated email responses and workflow execution
- Applied NLP concepts and intent-based processing techniques to improve conversational flow and user interaction handling
- Designed scalable user interaction flows and explored approaches for integrating advanced LLM-based conversational capabilities
- Rapidly prototyped and tested features to validate usability, performance, and business value
- Defined system requirements and user interaction flows to ensure a seamless user experience
- Collaborated with stakeholders to align the solution with business communication needs
- Documented system architecture, workflows, and implementation processes

Key Achievements:

- Delivered a functional AI chatbot capable of automating communication and workflow processes
- Improved operational efficiency by reducing manual communication tasks
- Demonstrated practical application of AI automation and conversational AI in solving business problems
- Successfully integrated multiple technologies into a cohesive solution

Junior AI Engineer | September 2020– December 2022**Loctect I.T Training Institute**

Nigeria

Fire and Smoke Detection System (Computer Vision)

Developed a computer vision-based AI system for early detection of fire and smoke to support safety monitoring and hazard prevention.

Key Responsibilities

- Designed and implemented a computer vision model for detecting fire and smoke in image and video data
- Collected, processed, and labelled datasets to train supervised machine learning models
- Applied image processing techniques and deep learning frameworks to improve detection accuracy
- Evaluated model performance using appropriate metrics and optimized for real-time detection
- Integrated the model into a prototype system for practical monitoring applications
- Conducted testing and validation to ensure reliability in different environmental conditions
- Documented system design, model architecture, and evaluation results

Key Achievements:

- Developed a functional fire and smoke detection system for early hazard identification
- Improved model accuracy through data preprocessing and iterative optimisation
- Demonstrated real-world application of AI in safety and risk management
- Delivered a scalable prototype suitable for integration into monitoring systems

TECHNOLOGIES AND TOOLS

Python, TensorFlow, PyTorch, NumPy, Pandas, Open AI, Claude, Gemini, Langchain, API Integration, AI POCs, AI Governance, OpenCV, REST APIs, Git, Docker, VS Code, SQL, HTML/CSS/JavaScript, AWS/Azure (Fundamentals), Conversational AI, NLP Concepts, Prompt Engineering, LangChain (Learning), Agentic AI Concepts, AI Workflow Automation, API-Based LLM Integration, Software Development,

EDUCATION AND CERTIFICATION

- Master's in Artificial Intelligence and Robotics, University of Hertfordshire, United Kingdom
- Member, British Computer Society (BCS)
- Bachelor's degree in Computer Science (B.Sc.), University of Port Harcourt, Nigeria